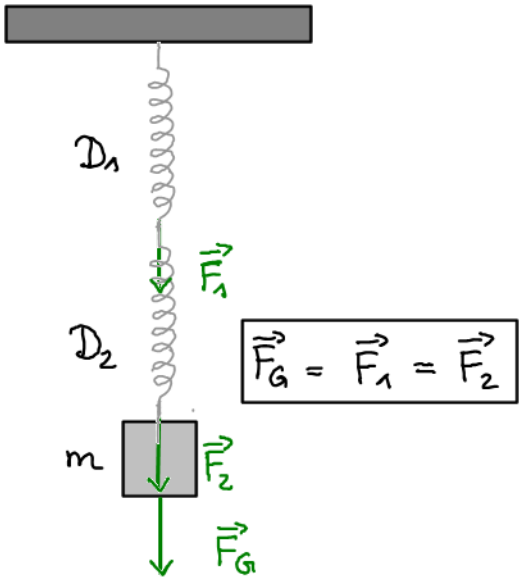
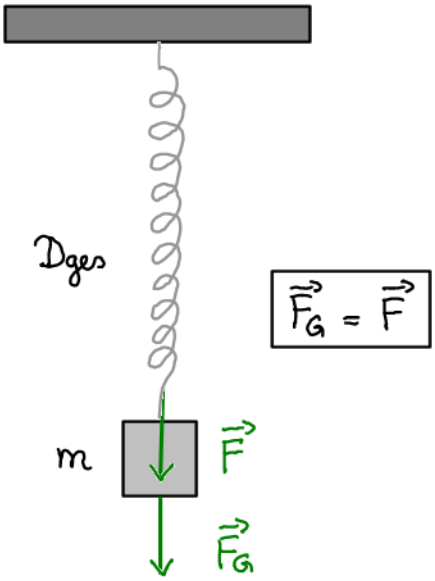
Lösung: Parallel oder in Reihe verbaute Federn

a) 2 Federn in Reihe



("Ersatzdarstellung")



Auslenkung:

$$\Delta s = \Delta s_1 + \Delta s_2$$

$$= \frac{F_1}{D_1} + \frac{F_2}{D_2} = F_G \cdot \left(\frac{1}{D_1} + \frac{1}{D_2}\right)$$

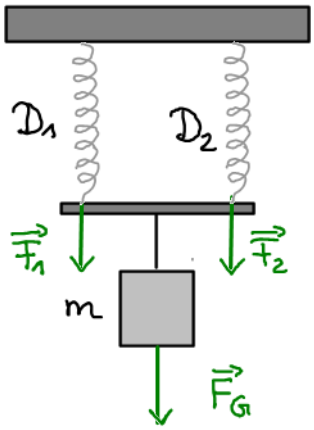
$$\Delta s = \frac{F}{D_{ges}} = \frac{1}{D_{ges}} \cdot F_G$$

Es kommt in den beiden verglichenen Systemen zur gleichen Auslenkung, wenn ...

$$F_G \cdot \left(\frac{1}{D_1} + \frac{1}{D_2}\right) = \Delta s = \frac{1}{D_{ges}} \cdot F_G$$

$$\Rightarrow \quad \boxed{\frac{1}{D_{ges}} = \frac{1}{D_1} + \frac{1}{D_2}}$$

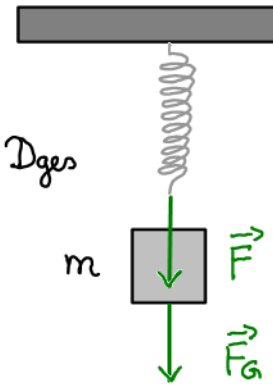
b) 2 parallele Federn



$\Delta s_1 = \Delta s_2$
gl. Auslenkung

$\vec{F}_G = \vec{F}_1 + \vec{F}_2$

("Ersatzdarstellung")



$\vec{F}_G = \vec{F}$

Kräfte:

$\vec{F}_G = \vec{F}_1 + \vec{F}_2$ $\qquad$ $\vec{F}_G = \vec{F}$

Es kommt in den beiden verglichenen Systemen zur gleichen Auslenkung $\Delta s = \Delta s_1 = \Delta s_2$, wenn..

$$\vec{F}_1 + \vec{F}_2 = F_G = F$$

$$D_1 \cdot \Delta s_1 + D_2 \Delta s_2 = D_{ges} \cdot \Delta s$$

$$D_1 \cdot \Delta s + D_2 \Delta s = D_{ges} \cdot \Delta s$$

$$\cancel{\Delta s}\ (D_1 + D_2) = \cancel{\Delta s} \cdot D_{ges}$$

$$\boxed{D_{ges} = D_1 + D_2}$$